INCH-POUND

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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

FILES, ROTARY, CYLINDRICAL SHAPE, BALL NOSE, HIGH SPEED STEEL OR TUNGSTEN CARBIDE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) establishes the government acquisition requirements for high speed steel or tungsten carbide, cylindrical shaped, ball nosed, rotary files.

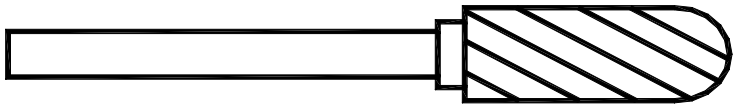


FIGURE 1. Cylindrical shape ball nose rotary file.

2. CLASSIFICATION. The rotary files shall be of the following types, styles, sizes, and classes:

2.1 Type. See 7.3(b)).

Type I - High speed steel
Type II - Tungsten carbide

2.2 Style. See 7.3(c)).

A - Right hand spiral fluting
B - Right hand spiral fluting with chip breaker
C - Double cut fluting
D - Diamond cut fluting

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: STDZNMGT@dla.mil or Defense Supply Center Richmond (DSCR), ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616.

AMSC N/A FSC 3455

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2.3 Sizes. The sizes for the rotary files shall be selected from tables I and II for the file specified (see 7.3(d)).

TABLE I. Type I high speed steel rotary files sizes and characteristics.

Size	Characteristics		
	Shank diameter (inches)	Cutter head dimensions (inches)	
		Diameter	Length
01	1/8	1/8	5/8
02	1/8	3/16	5/8
03	1/4	1/8	1/2
04	1/4	3/16	1/2
05	1/4	1/4	1
06	1/4	1/4	1-1/2
07	1/4	5/16	1
08	1/4	3/8	1
09	1/4	3/8	1-1/2
10	1/4	7/16	1
11	1/4	1/2	1
12	1/4	1/2	1-1/2
13	1/4	5/8	1
14	1/4	3/4	1-1/4
15	1/4	3/4	1-1/2
16	1/4	1	1-3/8

TABLE II. Type II tungsten carbide rotary files sizes and characteristics.

Size	Characteristics		
	Shank diameter (inches)	Cutter head dimensions (inches)	
		Diameter	Length
01	1/8	3/32	7/16
02	1/8	1/8	9/16
03	1/8	1/4	1/2
04	1/8	5/32	1/2
05	1/8	3/16	1/2
06	1/4	1/8	1/2
07	1/4	5/32	5/8
08	1/4	3/16	5/8
09	1/4	1/4	5/8
10	1/4	5/16	3/4
11	1/4	3/8	3/4
12	1/4	7/16	1
13	1/4	1/2	1
14	1/4	5/8	1
15	1/4	3/4	1
16	1/4	1	1

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2.4 Class. See 7.3(e)).

Class A - Coarse tooth spacing
Class B - Medium tooth spacing
Class C - Fine tooth spacing

3. SALIENT CHARACTERISTICS

3.1 Design. Type I and II rotary files shall be either one- or two-piece construction. When two-piece construction is used, the files shall have a steel shank securely joined to the cutting head in a manner that conforms to the manufacturer's standard commercial practice.

3.2 Materials and hardness.

3.2.1 Type I. The cutting heads of type I rotary files shall be manufactured from high speed tool steel in accordance with ASTM International A 600, "Standard Specification for Tool Steel High Speed". The cutting heads shall be hardened and tempered to a hardness of not less than 62 and not greater than 65 on the Rockwell C scale. Type I rotary files of two-piece construction shall have shanks of a hardness conforming to the manufacturer's standard part drawing for the rotary file specified in the acquisition order.

3.2.2 Type II. The cutting heads of type II rotary files shall be manufactured from C-2 commercial grade tungsten carbide. The cutting heads shall have a hardness of not less than 91.5 and not greater than 92.5 on the Rockwell A scale. Type II rotary files of two-piece construction shall have shanks hardened and tempered to a hardness of not less than 30 and not greater than 55 on the Rockwell C scale.

3.3 Head and shank concentricity. The head and shank shall be concentric to within 0.005 inch total indicator reading (TIR).

3.4 Tolerances. The allowable shank diameter tolerance shall be +0.000, -0.002 inch. The shank length shall conform to the manufacturer's standard part drawing for the rotary file specified in the acquisition order. The allowable tolerance for cutter head diameters shall be not greater than ±0.010 inch. The allowable tolerance for the cutter head length shall be not greater than ±0.062 inch.

3.5 Teeth per circumference. The number of cutting teeth on the circumference of the rotary file shall conform to table III or IV for the type rotary file and class tooth spacing required. The tooth count per circumference shall be the number of right hand spirals on the circumference of the rotary file. Where ranges are shown in tables III and IV for a particular diameter rotary file, the required number of teeth shall be not less than the lower number shown and not greater than the larger number shown.

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TABLE III. Number of teeth per circumference, type I rotary files.

Cutter head diameter (inches)	Number of teeth		
	Class A (coarse)	Class B (medium)	Class C (fine)
1/8	6 - 8	9 - 11	12 - 15
3/16	9 - 11	12 - 14	15 - 20
1/4	10 - 13	14 - 16	17 - 21
5/16	11 - 16	17 - 20	21 - 26
3/8	12 - 17	18 - 22	23 - 30
7/16	13 - 18	19 - 23	24 - 31
1/2	14 - 20	21 - 26	27 - 35
5/8	17 - 22	23 - 29	30 - 40
3/4	18 - 24	25 - 32	33 - 44
7/8	19 - 25	26 - 33	34 - 45
1	20 - 27	28 - 41	42 - 55

TABLE IV. Number of teeth per circumference, type II rotary files.

Cutter head diameter (inches)	Number of teeth		
	Class A (coarse)	Class B (medium)	Class C (fine)
1/8	8 - 11	12 - 15	16 - 20
3/16	10 - 12	13 - 16	17 - 24
1/4	12 - 15	16 - 20	21 - 26
5/16	13 - 16	17 - 22	23 - 30
3/8	13 - 18	19 - 26	27 - 38
7/16	15 - 20	21 - 27	28 - 36
1/2	16 - 22	23 - 28	29 - 39
5/8	18 - 25	26 - 28	39 - 47
3/4	19 - 28	29 - 38	39 - 49
7/8	22 - 29	30 - 37	38 - 55
1	24 - 32	33 - 43	44 - 62

3.6 Fluting. The fluting of the rotary file shall be either right hand spiral, right hand spiral with chipbreaker, double cut, or diamond cut, depending upon the style specified. The fluting shall create a cutting tooth pattern that extends along the entire periphery of the cutting head.

3.6.1 Style A fluting. The style A rotary file shall have right hand spiral fluting consisting of a standard commercial right hand helix. The helix angle, measured from the centerline of the cutter head, shall be not less than 15 degrees and not greater than 35 degrees.

3.6.2 Style B fluting. The style B rotary file shall have right hand spiral fluting with chipbreaker consisting of standard right hand spiral fluting with an additional series of slight serrations along the periphery of the cutting teeth that serve to break up the formation of long chips.

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3.6.3 Style C fluting. The style C rotary file shall have double cut fluting consisting of a series of left hand spirals in conjunction with the standard right hand spirals, which results in a wedge or chisel form cutting tooth. The number of left hand spirals shall be not less than 40 percent and not greater than 60 percent of the number of right hand spirals.

3.6.4 Style D fluting. The style D rotary file shall have diamond cut fluting consisting of a series of left hand spirals in conjunction with the standard right hand spirals, which results in a pyramidal form cutting tooth that may have either a sharp pointed top or a truncated top. The number of left hand spirals shall be not less than 70 percent and not greater than 90 percent of the number of right hand spirals.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

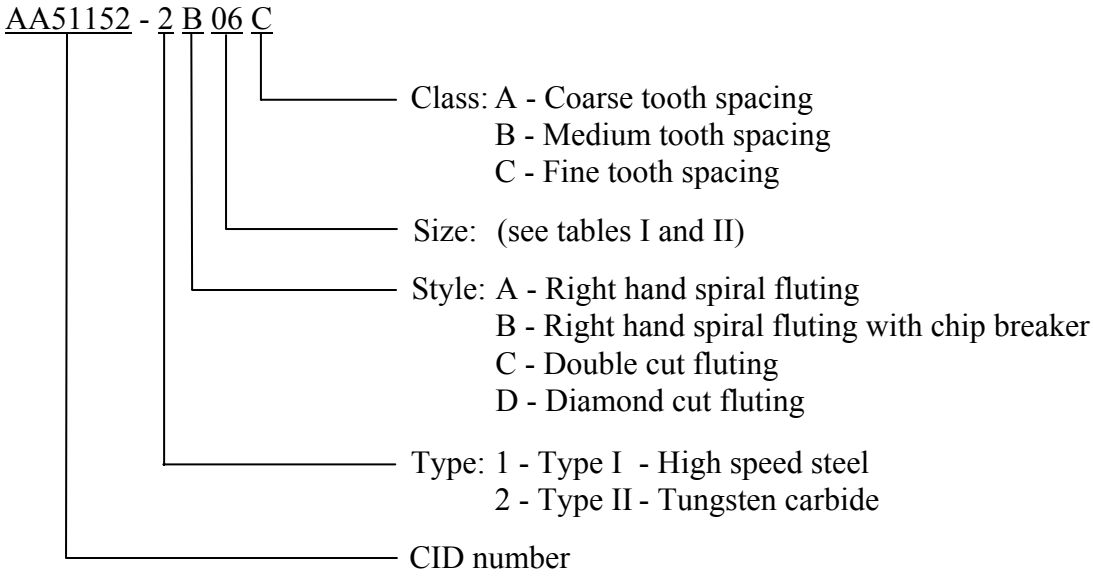
6. PACKAGING

6.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the acquisition order (see 7.3(f)).

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7. NOTES

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



7.2 Sources of documents.

7.2.1 FAR. The FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of FAR documents may be obtained from http://www.arnet.gov/far/.

7.2.2 ASTM standards. Copies of ASTM standards may be obtained from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959. Electronic copies of ASTM standards may be obtained from http://www.astm.org/.

7.3 Ordering data. The acquisition order should specify the following information:

a. CID document number, revision, and CID PIN.
b. Type (see 2.1).
c. Style (see 2.2).
d. Size (see 2.3).
e. Class (2.4).
f. Packaging (see 6.1).

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7.4 Subject term (key word) listing.

chip breaker
cutting head
diamond cut
double cut
fluting
spiral

MILITARY INTERESTS:

Custodians:
Navy - SH
Air Force - 99

Review Activity:
Air Force - 84

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - FSS

Preparing Activity:
DLA - GS6

(Project 3455-0182)

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